MILITARY SPECIFICATION

# SCALES, GRAPHICAL FIRING, GENERAL SPECIFICATION FOR

This specification is approved for use by the US Army Armament Research and Development command, and is available for use by all Deparments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers three groups of graphical firing scales identified herein as Type 1, Type 2 and Type 3. The scales provide ballistic data and determine corrections pertinent to the firing of artillery weapons.

### 1.2 Classification

1.2.1 Type 1 Scales. This type is identified as a graphical firing scale which consists of a rule and an indicator assembly with hairline.

1.2.2 Type 2 scales. This type is identified as a graphical site scale which consists of a rule, and indicator assembly with hairline and a slide.

1.2.3 Type 3 scales. This type is identified as a graphical munitions effect scale and consists of a rule and indicator assembly.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form part of this specification to the extent specified herein.

*U.S. GOVERNMENT PRINTING OFFICE 1983-605-034/4562

FSC: 1220

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-F-13926 | Fire Control Materiel: General Specification governing the Manufacturing and Inspection of. |
| MIL-I-45607 | Inspection Equipment, Supply and Maintenance of |

STANDARDS

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for inspection by Attributes. |
| MIL-STD-109 | Quality Assurance Terms and Definitions |
| MIL-STD-810 | Environmental Test Methods |

DRAWINGS

US ARMY ARMAMENT RESEARCH(2 AND DEVELOPMENT COMMAND
as specified in specification sheets
(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Specifiations sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern.

3.2 Materials. Materials shall be in accordance with drawings, material specifications, and general specifications forming part of this specification.

3.3 Fabrication. The scales shall be manufactured in accordance with the applicable drawing listed in the appropriate specification sheet and all drawings pertaining thereto.

3.4 General specifications. The contractor shall be responsible for adherence to and compliance with the following requirements of MIL-F-13926.

a. Order of precedence.
b. Dimensions and tolerances.
c. Part identification and marking.
d. Workmanship.



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3.5 Environmental.

3.5.1 Storage Temperatures. The scales shall meet the requirements of 3.6 when thermally stabilized at the standard ambient temperature of between 60° and 90°F after having been exposed to and thermally stabilized at ambient temperature of -80° and +160°F.

3.5.2 Operating Temperatures. The scales shall meet the requirements of 3.11 when exposed to and thermally stabilized at ambient temperatures of -40° and +150°F.

3.5.3 Humidity. The scales shall meet the humidity requirements of MIL-STD-810, Procedure IV. The scalels shall then be thermally stabilized after return to the standard ambient temperature of between 60° and 90°F, and Shall meet the requirements of 3.6.

3.6 Post-Environmental. Subsequent to the temperature and humidity tests specified in 3.5.1 and 3.5.3 the scales shall be operable and shall show no color changes, distortion, warpage, breaks, stickiness, cracks, mottling, or separation of bonded materials; and the scales shall meet all of the requirements of 3.7 to 3.11 inclusive.

3.7 Printing and surface quality. Letters and numbers shall be in accordance with the applicable drawings. They shall be sharp, evenly colored, without skips or breaks, and shall not touch each other or any line of the scale. The printing surface shall be free from stains or imperfections.

3.8 Scales. All scales shall be printed in the colors noted on pertinent drawings. The graduations shall be accurately placed so that error will not affect result of camputations listed in 3.10 in excess of the tolerances indicated.

3.9 "C" and "D" SCALES GRADUATION ERRORS - Type 2 scales.

3.9.1 With the first graduation of the “C” scale in coincidence with the first graduation of the "D" scale, each succeeding graduation on the "C" scale shall be in alignment with the respective graduation on the "D" scale within 0.005 inch.

3.9.2 A 2/3 ratio between graduations on the “C” and “D” scales shall be read throughout the scale within the tolerance indicated in 3.9.1 when the first graduation on the “C” scale is set in coincidence with the 1.5 graduation on the "D” scale.

3.10 Accuracy



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3.10.1 Type 1 and Type 2 Scales. Scale readings indicated in the appropriate specification sheet shall be obtained within the tolerances specified, for each respective range and condition listed.

3.11 Operability. The indicator and the slide for type 2 scale shall move freely in relation to the rule when the scales are exposed to, and thermally stabilized at the standard ambient temperature of 3.5.1, the extreme operating temperatures of 3.5.2, and the humidity conditions as specified in 3.5.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibiliy for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection and test requirements as specified herein. Except as otherwise specified, in the contract or purchase order the supplier may utilize his own facilities or any other facility acceptable to the government for the performance of the inspection and test requirements stated herein. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are-deemed necessary to assure-supplies and services conform to prescribed requirements.

4.1.1 General Provisions. The provisions and requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First Article Provisions

4.2.1 First Article Submission. This specification contains technical provisions for first article inspections and tests. Requirements for the submission of first article samples by the contractor shall be as specified in the contract and as stipulated by the Contracting Officer.

4.2.2 First Article Approval. The requirement for first article approval and the responsibility for first article testing and inspection shall be as specified in the contract. The sample for first article shall consist of 3 graphical firing scales manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.3 Contractor Testing. When the contractor is responsible for conducting first article approval inspections and tests, the sample shall be inspected by the contractor for all the



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requirements of the contract. The sample and record of this inspection, including certificates of conformance for raw and finished material, shall be submitted to the government for approval. The government reserves the right to witness the contractor's inspection.

4.2.4 First Article Inspections and tests. Unless otherwise specified in the contract or purchase order or stipulated by the Contracting Officer, the first article sample shall be subjected to all inspections and tests stated or referenced herein, and to all requirements of the applicable drawings.

4.2.5 First Article Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly, component, or test specimen in the sample, to comply with any of the stated requirements.

4.3 Inspection Provisions.

4.3.1 Submission of Product. Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" Provisions of MIL-STD-105. Unless otherwise specified by the Contracting Officer, any material found defective shall be removed from lots and shall not be submitted to the Government for acceptance.

4.3.2 Examination and Tests. Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP) and those stated herein. In the absence of QAP's, absence of the applicable Quality Assurance of MIL-F-13926 shall apply. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105, and the sample plans specified in Table I and Table II herein. Examinations and tests for Packaging, Packing, and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to government acceptance or rejection by lot.



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TABLE I. CLASSIFICATIION OF DEFECTS

| CLASS | CHARACTERISTIC | REQUIREMENT | TEST PROCEDURE |
|---|---|---|---|
| CRITICAL: NONE | | | |
| MAJOR: AQL 1.0 PERCENT DEFECTIVE | | | |
| 101. | Printing and Surface quality | 3.7 | 4.6.6 |
| 102. | Scales | 3.8 | 4.6.7 |
| 103. | Graduation errors | 3.9 | 4.6.8<br>4.6.8.1 |
| 104. | Accuracy | 3.10 | 4.6.9<br>4.6.9.1<br>4.6.9.2 |
| 105. | Operability | 3.11 | 4.6.10 |
| MINOR: NONE | | | |

4.3.3 Acceptance and Rejection. Rejected lots shall be screened for all defect characteristics. Removal for correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special Sampling. Three graphical firing scales shall be selected at random from each 50 produced or from each month, whichever occurs first. The special samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests in table II

TABLE II

| NO. | CHARACTERISTIC | REQUIREMENT | TEST PROCEDURE |
|---|---|---|---|
| 301 | Fabrication | 3.3 | 4.6.1 |
| 302 | General Specification | 3.4 c & d | Visual, MIL-F-13926 |
| 303 | Storage Temperature | 3.5.1 | 4.6.2 |
| 304 | Operating Temperature | 3.5.2 | 4.6.3 |
| 305 | Humidity | 3.5.3 | 4.6.4 |

4.4.1 Failure of Sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection Equipment. Except as otherwise provided for by the contract, the contractor shall supply equipment to perform all the inspections and tests referenced and as stated in the contract.



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Such equipment shall be in accordamce with the applicable requirements of MIL-I-45607. The government reserves the right to use the test equipment for its own independent inspectionss to the extent that such use will not unduly interfere with thew contractor's delivery schedule.

4.5.1 Government Furnished Inspection Equipment. Where the contract provides for government test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor Furnished Inspection Equipmemt

4.5.2.1 Government Design. All inspection equipment specified by drawing number in specifications or QAP's forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the Technical Data Package List.

4.5.2.2 Contractor Designs - Contractor designs are required for all inspection equipment for which government design are not specified and may include commercial equipment which the contractor proposes to use. Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public. The contractor snail design and supply inspection and test equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test facilities" requirements of MIL-F-13926 as applicable. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facili routine calibration of test equipment. Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commerical equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture (s) required to be used with commercial equipment, or with Government designs if not otherwise revered thereby. Designs shall be of the category and form specified in the Contract Data Requirements Lists. The contractor is referred to MIL-HDBK-204 "Inspection Equipment Design" for guidance. The item detail



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specification number, paragraph number, and defect number from Section 4 shall be referenced on each contractor design together with the component or assembly drawing number, revision letter and date to which the specific design applies.

4.5.2.3 Submission of Designs for Approval - Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Design shall be submitted for approval in accordance witin the stipulations, time frame and distribution specified in the Contract Data Requirements Lists. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. The specific segment of the Government to which the Contractor designs shall be sent will be specified in the contract.

4.6 Test Methods and procedures

4.6.1 Fabrication. The graphical firing scales shall be subjected to a visual, tactile, and dimensional inspection for conformance to the applicable drawings and requirements of this specification. Dimensional inpection shall be performed using standard measuring equipment.

4.6.2 Storage Temperatures. The scales shall be exposed to the temperatures specified in 3.5.1, utilizing testing equipment conforming to the “Test Facilities” requirements of MIL-STD-810. The equipment shall be capable of maintaining the specified temperatures until thermal stabilization is reached. The temperature shall be changed gradually to avoid thermal shock. The scales shall lay on a flat surface in the test chamber, and the flat surface shall support the entire length and width of each scale. The scales shall be exposed to each specified temperature for a period of five hours. After being exposed to the above conditions, and upon return to the standard ambient temperature of 3.5.1, the scales shall be subjected to the test in 4.6.5 and shall conform to the requirements specified.

4.6.3 Operating Temperature. The scales shall be exposed to the temperatures specified in 3.5.2 utilizing testing equipment conforming with the “Test Facilities” requirements of MIL-STD-810. The equipment shall be capable of maintaining the specified temperatures until thermal stabilization is reached. The temperatures shall be changed gradually to avoid thermal shock. When thermal stabilization is reached at each temperature extreme, the temperature shall be maintained constant for a minimum of five hours for each condition. At the end of each time period and while still at the temperature extreme, the scale shall be tested in accordance with 4.6.10 to determine operability in compliance with 3.11 as required by 3.5.2.



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4.6.4 Humidity. This test shall be conducted in accordance with method 507.1, procedure IV defined in MIL-STD-810. During each test measurement period shown on figure 507.1-3, the scale shall be inspected for physical damage and shall be tested in accordance with 4.6.10 to determine operability in compliance with 3.11. Upon completion of this test, the scale shall be subjected to the tests specified in 4.6.5 and shall conform to the requirements referenced therein.

4.6.5 Post-environmental. Subsequent to the tests specified in 4.6.2 and 4.6.4 the scale shall be subjected to a visual and tactile examination for defects or damage as specified in 3.6. The scale shall then be subjected to the tests specified in 4.6.6 to 4.6.10 inclusive, and shall conform to the requirements referenced therein.

4.6.6 Printing and surface quality. This test consists of a visual examination of the face of the scale for imperfections of coloring applications, letters, numbers, surface quality, stains and other imperfections in accordance with the requirements of 3.7.

4.6.7 Scales. All scales shall be visually inspected for quality of printed graduations and to insure that colors are in accordance with pertinent drawings in compliance with the requirements of 3.8. The graduations shall then be inspected for accuracy of divisions and subdivisions by being subjected to the test specified in 4.6.3 and shall conform to the requirements specified.

4.6.8 Graduation error - “C” and “D” scales. Determine graduation error between the “C” and "D" scales by utilizing standard measuring equipment capable of measuring any misalignment of the scale graduations within the tolerance specified. Move the center slide until the index of the "C" scale is in coincidence with the index of the “D” scale and clamp the slide and the rule. together lightly to prevent movement. Each graduation on the "C" scale shall be checked for alignment with the opposite graduation on the “D” scale within the tolerance specified in 3.9.1.

4.6.8.1 Accuracy. - ratio “C” and “D” scales. To determine the ratio of 2/3 for the "C" scale with respect to the "D" scales, utilize the same set up as specified in 4.6.8. Align the index of the “C” scale opposite the "15" graduation line of the "D" scale and clamp the slide and rule together lightly to prevent movement. At this setting, a ratio of 2/3 shall be read throughout the scale with the opposite graduations in alignment within the tolerance specified in 3.9.2.



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4.6.9 Accuracy

4.6.9.1 Type I. The test for accuracy shall be visually conducted with the aid of an optical device with a magnification of at least 10 power. The accuracy of the scale shall be checked against the respective table specified on the applicable specification sheet. Superimpose the hairline of the indicator assembly over the specified range line, and read the value on the scale. The readings noted shall be accurate within the tolerances specified in 3.10.1

4.6.9.2 Type II. The test for accuracy shall be visually conducted utilizing an optical device as in 4.6.9.1. Position the hairline of the indicator assembly over the specific vertical interval on the “D” scale indicated in the respective table specified on the applicable specification sheet. The center slide shall then be moved until the selected range is under the hairline of the indicator assembly. The scale shall be held in this position by applying pressure with the thumb and forefinger of one hand, and with the other hand move the hairline of the indicator assembly to the specified gage point and read the numerical value on the “D” scale. The readings shall be accurate within tolerances specified in 3.10.2.

4.6.10 Operability. While stabilized at each temperature as specified in 3.5.1, and 3.5.3 the scale shall move freely when normal thumb pressure is applied to move the indicator assembly and the slide for type 2 scale. The indicator assembly and the slide for type 2 scale shall be moved several times back and forth to determine conformance with the requirement of 3.11.

5. PACKAGING

5.1 Packaging, Packing, and Marking. Packaging, packing and marking shall be in accordance with MIL-P-14232. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.

6.1.1 Type 1 scales. The graphical firing scales are used to provide ballistic data pertinent to the application of weapon fire for a particular range and condition.

6.1.2 Type 2 scales. The graphical firing scales are used to determine corrections to quadrant elevation. Correction readings are determined in accordance with the "Instructions on the reverse side of the scales.”



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6.1.3 Type 3 scales. The graphical munitions effect scales are used to provide statistical data for munitions effect based upon ammunition calibers and target types.

6.2 Ordering data. Procurement documents should specify the following.

a. Title, number, and date of this specification.

b. Type of scales with reference to applicable specification sheet.

c. Selection of applicable levels of preservation, packaging, and packing.

d. Applicable stock number.

Custodian:
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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-S-64045(AR) | SCALE, GRAPHICAL FIRING, GENERAL SPECIFICATION FOR |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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*NOTE:* This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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